



Group 56 Flight Report

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Summary of flight performance

For this assignment, two test flights were undertaken. The first flight was done to make sure the glider could fly and to make note of any necessary improvements to the glider's design. The glider crashed during the first flight and several components needed to be repaired. The second flight was done after changes to the glider's design were made and allowed for measurements to be taken so that a complete analysis of the flight could be made.

Results

Table1: Table showing the measured values of voltage, current, revolutions per minute and analysis of the stability of the glider during its flight.

Measurement	Value	Observations
Voltage (Constant)	13V	Supplied via bench power supply.
Current (Take-off)	3.2 A	The glider had a fast response time and took off quickly without stalling.
Current (Cruise)	2.1 A	The glider was flying steadily and was stable.
Revolutions per minute (Controlled flight)	24.4	There were no oscillations.
Stability of glider	Stable at high speeds but with a few oscillations at low speeds.	At low speeds there was a bit of pitch instability which caused the glider to oscillate, but at high speeds the glider was very stable in the pitch direction.

Observations and Improvements

During the second flight test, the variable bench power supply maintained a constant voltage of 13.0V. As a result, the electrical power consumed by the system ($P = IV$) was directly proportional to the current draw, as it dropped from 41.6 W (3.2A) during takeoff to 27.3W (2.1A) during cruise. This large power reduction indicated that the glider successfully transitioned from acceleration into cruise.

High initial power was required to overcome the inertia of the stationary glider and the increased induced drag of the glider during take-off. Once the aircraft transitioned into a steady cruise the drop in power consumption indicated an optimised lift-to-drag ratio, which allowed the glider to maintain a stable horizontal flight path.

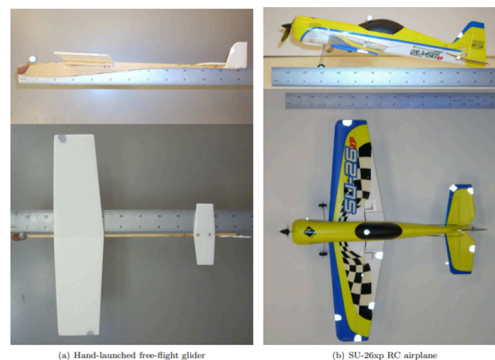
To estimate the aerodynamic efficiency of the glider the TAS was calculated from the angular velocity recorded (24.4RPM), resulting in the cruise velocity being 5.88ms^{-1} . In level flight, the lift was balanced by the weight of the glider. Assuming a mass of around 150g, the lift was approximately equal to 1.47N. By using empirical data on glider testing (ResearchGate, 2025), it was estimated that the glider had a L/D ratio of approximately 2.2. Using this and the estimated lift produced, the drag at cruise speed was approximated to 0.67N. Using this and the TAS (5.88ms^{-1}) the aerodynamic power output could be estimated to be approximately 3.94W ($P = D \times TAS$).

Since the electrical power input was 27.3W, the total propulsion efficiency was calculated to be approximately 14.4% (see Appendix 2 for full calculation). The total propulsion efficiency of 14.4%, whilst seemingly low, was the compounded electrical motor loss and propeller aerodynamic inefficiency. Whilst the electric motor likely had a high efficiency, the low Reynolds number increased the relative viscosity of air which led to flow separation and thick boundary layers along the blades, hindering propeller's thrust capability (Brandt and Selig, 2011). Additionally, the propeller was not optimised for the glider's exact cruise advance ratio and, as a result, most of the electrical power input was likely lost overcoming the profile drag of the blades and the parasitic drag of the rapidly rotating tether line.

When designing the glider, the main aim was to achieve as much lift as possible and make the glider as light as possible. As a result, the design of the tailplane was not prioritised which affected the performance of the glider. This was observed during the first test flight, as at low speeds ($\sim 1.7 \text{ A}$) the glider was unstable and produced oscillatory motion which resulted in it crashing into the ground. This was due to the glider being unable to respond to small adjustments in speed which led to low precision in the pitch control (Drela, M, 2006).

This was improved by adjusting the shape of the horizontal stabiliser. After conducting some research and focusing on designs that had high success rates, it was decided that its shape should be similar to that of a parallelogram, as shown in Figure 1.

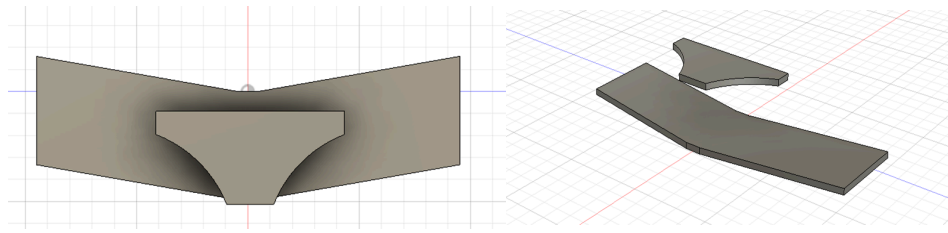
Figure 1 showing examples of gliders with horizontal stabilisers in the shape of a parallelogram:



In order to achieve maximum stability the shape of the stabiliser needed to be optimised mathematically. The calculation for this is shown in Appendix 1.

After finding the dimensions to optimise the shape of the tailplane, a CAD design was created as shown in the figure below.

Figure 2 showing the optimised CAD design of the tailplane



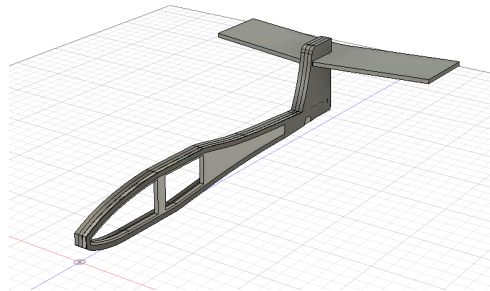
The new design was significantly larger than the previous one and proportional to the size of the glider which helped to provide better pitch stability. (TU Delft).

Another improvement made to the glider was to add a T-tail. This was done because in the previous design, the horizontal stabiliser was mounted very low on the tailplane, which meant that it was located in the downwash generated from the wings. This was a problem as it caused turbulent air to flow over the stabiliser, creating a loss in lift. As the glider had minimal control surfaces, this also contributed to the oscillatory motion. (Perkins, C.D and Hage, R.E, 1949)

Therefore, placing the stabiliser in a higher location meant that it would no longer be in the wing's wake, allowing it to increase the glider's stability.

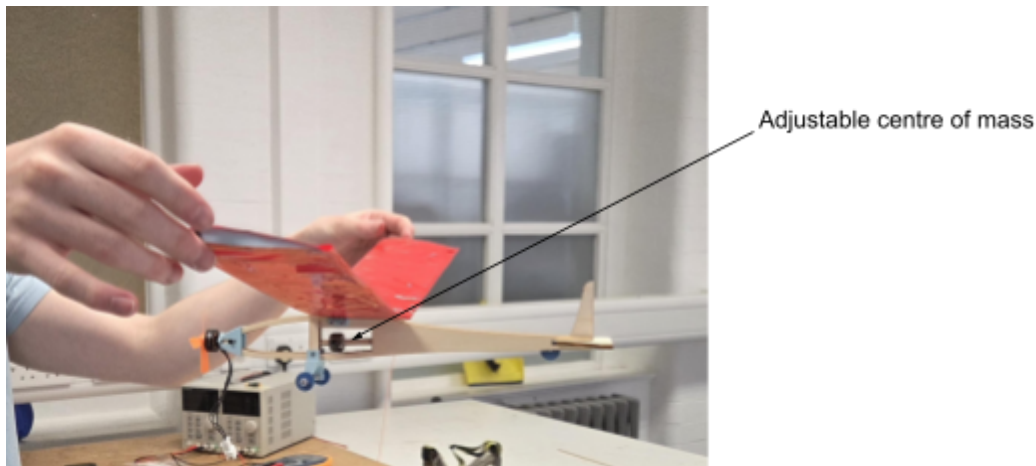
However, moving the stabiliser to a higher position meant that the vertical stabiliser would need to be thicker and longer to support the weight of the stabiliser. This was taken into account when redesigning the rear empennage as shown in the figure below.

Figure 3 showing the rear empennage CAD design.



Another important observation was that at high speeds ($\sim 3.2 \text{ A}$), the glider had a large amount of lift and flew at a steady level. This was due to the glider's centre of mass being placed in a suitable position. This was done by adding an adjustable centre of mass to the glider. The best position to place the centre of mass was determined by making the glider slightly nose heavy as shown in the figure below.

Figure 4 showing the location of the adjustable centre of mass

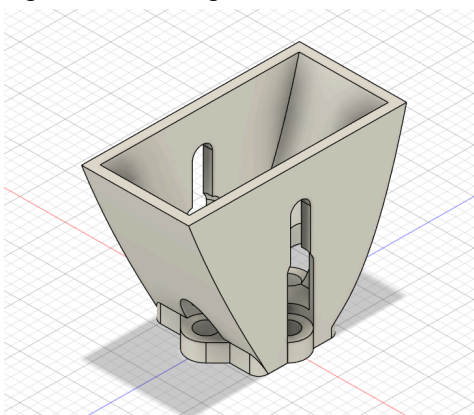


This greatly benefited the design of the glider as it allowed for improvements to be made. For example, the addition of the T-tail would have shifted the glider's centre of mass towards the tail, which would have affected its lift and caused it to have pitch instability. Therefore, to avoid this, the centre of mass was made adjustable, allowing the glider to benefit from the addition of the T-tail without compromising lift.

During the first test flight, oscillations caused by the aircraft's inherent instability resulted in a crash that broke the motor mount assembly. For the second flight the design of the motor mount was improved in order to strengthen it, and fix a clearance issue that prevented two of the motor attachment bolts from fitting.

The redesigned motor mount featured a reinforced forward bulkhead, and widened horizontal clearance channels as shown in the figure below. In addition to these changes, the decision to replace the 3D printed transverse bar with a single m3 nut and bolt was made during the build sessions. These modifications were done in order to improve the motors strength and increase its likelihood of surviving a crash as well as making the install process quicker and more simple as no filling was required

Figure 5 showing a CAD model of the redesigned motor mount



References

Drela, M. (2006). Lab 8 Notes – Basic Aircraft Design Rules. [online] MIT OpenCourseWare. Available at: https://ocw.mit.edu/courses/16-01-unified-engineering-i-ii-iii-iv-fall-2005-spring-2006/ffbff8a4985ee6ea819afb90c40834e3_spl8.pdf

Uhlig, D.V. and Selig, M.S. (2011). Stability and Control Characteristics of Micro Air Vehicles from Experimental Measurements. [online] illinois.edu. Available at: <https://m-selig.ae.illinois.edu/pubs/UhligSelig-2011-AIAA-2011-3820-MAV-Stability.pdf>

Merski, J. (1998). A Method for the Analysis of Longitudinal Static Stability of a Flight Vehicle. [online] Wright-Patterson Air Force Base, Ohio: Air Force Institute of Technology. Available at: <https://apps.dtic.mil/sti/pdfs/ADA361001.pdf>.

Raymer, D.P. (1992). Aircraft Design: A Conceptual Approach. 2nd ed. Washington, D.C.: American Institute of Aeronautics and Astronautics, pp. 76. Available at: http://deadnet.se:8080/Books_and_manuals_on_various_stuff/Science/Physics/Aircraft%20Design%20-%20A%20Conceptual%20Approach%202nd%20ed.%20-%20D.%20Raymer%20%281992%29%20WW.pdf

TU Delft (n.d.). Longitudinal Static Stability. [online] tudelft.nl. Available at: https://ocw.tudelft.nl/wp-content/uploads/Hand-out-Stability_01.pdf

Perkins, C.D. and Hage, R.E. (1949). Airplane Performance, Stability and Control. [online] New York: John Wiley & Sons. Available at: <https://archive.org/details/airplaneperforma0000perk>

ResearchGate. (2025). Effect of Wing Aspect Ratio on the Aerodynamic Performance and Stability of a Balsa Wood Glider. Journal of Applied Aerodynamics, 14(2), 112-121. https://www.researchgate.net/publication/399198626_Effect_of_Wing_Aspect_Ratio_on_the_Aerodynamic_Performance_and_Stability_of_a_Balsa_Wood_Glider?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Ii9kaXJlY3QiLCJwYWdlIjoicHVibGlicXlYXRpb24ifX0

Brandt, J.B. and Selig, M.S. (2011). Propeller Performance Data at Low Reynolds Numbers. 49th AIAA Aerospace Sciences Meeting, AIAA Paper 2011-1255.

Appendix 1

In order to achieve maximum stability the shape of the stabiliser needed to be optimised mathematically. This was done by using the equation shown below.

$$S_h = V_h \cdot S_w \cdot \frac{c}{L_h} \quad [1]$$

S_h = Tail Area (mm²)

V_h = Volume Tail Coefficient (no units)

S_w = Wing Area (mm²)

c = Chord

L_h = Tail Moment Arm

The values of the wing area and chord used were taken from the glider design, the tail moment arm was assumed to be 195mm. This was justified from research which showed that the centre of mass should be $\frac{1}{4}$ of the length down from the nose. The length of the glider was 284.2mm, hence the centre of mass is located 213.15mm from the nose, as shown from the calculation below:

$$284.2 - (284.2 \times 0.25) = 213.15 \quad [2]$$

Another assumption used was that the aerodynamic centre of the stabiliser was 20mm, measured from the back of the glider, leading to L_h being approximately 195mm for ease of design and calculation.

Research showed that the range for V_h should be between 0.3 and 0.6, however for the glider, 0.55 was chosen. This value was chosen as the oscillations observed in the first test flight were large, so a high volume tail coefficient would increase the static margin and reduce the glider's oscillations (Merski, J, 1998).

Inputting all the values into equation [1] resulted in the tail area being 9491mm.

To find the span of the stabiliser and the chord, the aspect ratio was needed. The recommended range for the aspect ratio was between 3 and 5 (Raymer, D.P, 1992), so an aspect ratio of 4 was chosen as it is the average. This resulted in the span being 195mm and the chord length being 49mm.

Appendix 2 - Efficiency Calculations

- Initial Parameters & Constants
 - Mass (m): 0.15kg
 - Angular Velocity (N): 24.4 RPM
 - Flight radius (r): 2.3m
 - Supply voltage (V): 13 V
 - Current at cruise (I_{cruise}): 2.1 A
 - Estimated Lift-to-drag-ratio (L/D): 2.2 (Source: ResearchGate, 2025)
 - Acceleration due to gravity (g): 9.81ms^{-2}

To calculate the True Airspeed (TAS) from the recorded angular velocity (N), the RPM was first converted to angular velocity in radians per second (ω).

$$\omega = N \times \frac{2\pi}{60} = 24.4\text{RPM} \times \frac{2\pi}{60} = 2.555\text{rad s}^{-1}$$

$$TAS = \omega r = 2.555\text{rad s}^{-1} \times 2.3\text{m} \approx 5.88\text{ms}^{-1}$$

The glider was in a steady flight which meant that the lift (L) produced by the glider was balanced by its weight (W).

$$L = W = mg$$

$$L = 0.150\text{ kg} \times 9.81\text{ ms}^{-2}$$

$$L = 1.47\text{N}$$

Using the estimated lift-to-drag ratio of 2.2, the total drag (D) at cruise speed was calculated.

$$D = \frac{L}{L/D} = \frac{1.47\text{N}}{2.2} \approx 0.67\text{N}$$

The aerodynamic power output (P_{out}) required to maintain the cruise velocity (TAS) is the product of the drag force and the velocity.

$$P_{out} = D \times TAS = 0.67\text{N} \times 5.88\text{m/s}$$

$$P_{out} \approx 3.94\text{W}$$

The total propulsion efficiency (η) is the ratio of the useful aerodynamic power output (P_{out}), to the total electrical power input ($P_{in} = V \times I_{cruise}$), expressed as a percentage as shown below.

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{P_{out}}{V \times I_{cruise}} \times 100 = \frac{3.94W}{2.1A \times 13V} \times 100$$

$$\eta \approx 14.4\%$$